STATS 298 Final Report — Industrial Research for Statisticians

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Team: Data & Analytics
Term: Summer 2025 (CPT)
Location: Austin, TX
Report Date: August 9, 2025

Overview

During Summer 2025 I completed an off-campus internship at Zello, a software company whose push-to-talk platform is used by frontline teams in construction, logistics, and public safety. Under the Data & Analytics organization, my work spanned applied machine learning for message understanding, analytics engineering with dbt, and operational efficiency. The internship provided substantive industrial research experience that built directly on my M.S. Statistics training and prepared me for full-time data science roles.

Project 1 — LLM Conversation Disentanglement

Zello's "AI Insights" feature summarizes long streams of voice/text messages. A prerequisite to accurate summaries is **conversation disentanglement**—partitioning interleaved messages into coherent threads. I designed and evaluated an LLM-based pipeline to assign conversation IDs to messages, focusing on construction-site communications.

- **Data & Privacy:** From real conversation logs, I created a **synthetic dataset** that preserved conversational structure while anonymizing content. This enabled safe iteration and reproducible evaluation.
- **Modeling:** I implemented a windowed prompting strategy with consistency checks across overlapping spans, plus simple graph-based post-processing to merge fragments.
- **Evaluation:** Using Adjusted Rand Index (ARI) as the primary metric, I improved performance from **0.31 to 0.92** on the synthetic test set, a level sufficient to unlock materially better downstream summaries and key-info extraction.
- **Statistical perspective:** The work emphasized metric selection, bias/variance trade-offs in prompt design, and robust validation under distribution shift between synthetic and live traffic.

Project 2 — Analytics Engineering with dbt

I contributed to Zello's analytics stack by building and hardening **dbt** models that serve product analytics and business reporting.

- **Deliverables:** Source models with tests and documentation, intermediate marts for feature usage and account-level health, and final models used by downstream dashboards.
- **Key learning:** Balancing **generalizable, reusable** models with **analysis-ready** outputs. I practiced modeling conventions (naming, sources, tests, freshness), incremental strategies, snapshotting, and dependency management so that analyses remain reliable as schemas evolve—an area difficult to master purely in coursework.

Project 3 — Operational Efficiency

To streamline go-to-market workflows, I built an automated **message pipeline** connecting **HubSpot** → **(LLM enrichment)** → **Slack** via Pipedream.

- The pipeline summarizes calls/emails/meetings, extracts customer-behavior signals (e.g., adoption blockers, feature interest), and posts actionable notes to the relevant Slack channels.
- This reduced manual transcription and improved the feedback loop between Customer Success and Product, illustrating how lightweight automation plus language models can create immediate business value.

Outcomes and Skills Gained

- **Technical:** Prompt engineering and evaluation for LLMs; conversation clustering; experiment design; dbt modeling (tests, documentation, CI-friendly patterns); SQL and data warehousing; API integration and event-driven automation.
- **Professional:** Cross-functional collaboration with Product Analytics and GTM teams; writing clear technical documentation; scoping MVPs and iterating with stakeholders.

Reflection

This internship let me experience the **day-to-day of data analysts and data engineers** working together to ship ML-powered features. It strengthened my foundations in measurement, reproducibility, and production-minded analytics engineering—competencies that complement my academic training in the M.S. Statistics program. This experience positions me well to pursue a full-time role in data science and to continue developing statistically sound, maintainable, and business-impactful solutions.